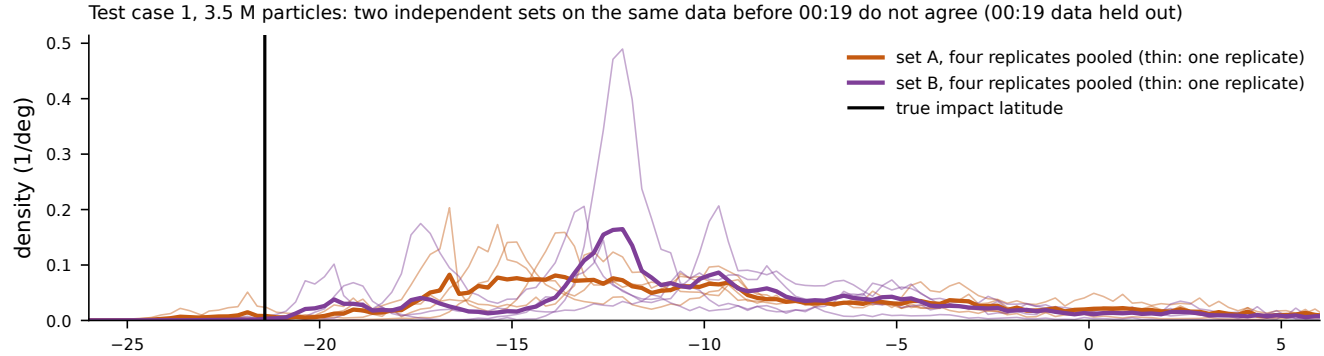
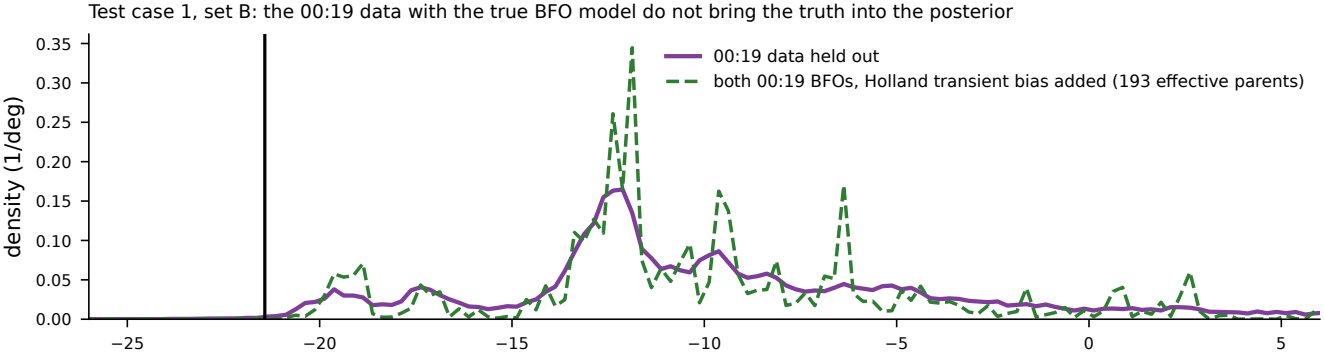
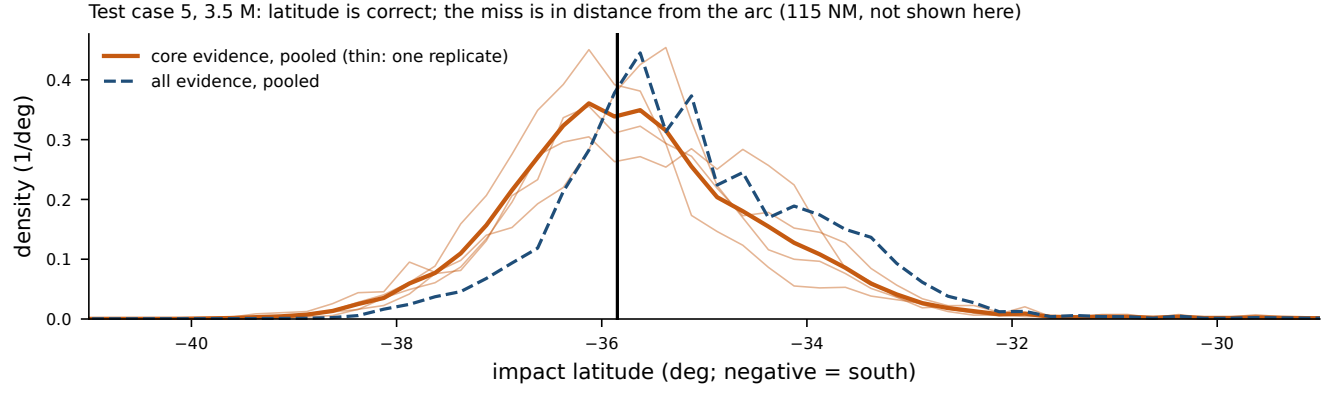
Posterior impact latitude at 3.5 million particles per replicate, test cases 1 and 5







Note (simplified English): black line = known true impact. Test case 1 was run twice with different random numbers; the two pooled curves differ, so the sampler is not converged for this case at 3.5 million particles. With the 00:19 data scored under the true BFO model the posterior stays away from the truth and has few effective samples. Test case 5 is correct in latitude. Physics of the end of flight is provisional.

Technical: waypoints stratum only; 4 replicates × 3.5 M particles (1/2 of run J production), 2×10^4 hand-off rows, 64 descents each; end of flight production recipe (round-3 laws, bank limit U[52, 90] deg, dive branch, C2-silent aim); log-on cause mixture q = 1/3, survival factor. Test case 1 truth: 00:19 BFO model start-up offset (set B data; set A used the inflated draw at 00:19 only, D0 data identical). All evidence = core + drift (segment level, GLORYS12 and GlobCurrent equal weight) + seabed searches (V1 central at the impact point, no settling, Bluefin-21 omitted). Hydroacoustics and Pléiades excluded. Smoke only.